

ORD SHARP, NE, USA

WMO: 725524

Lat: 41.623N

Lon: 98.948W

Elev: 2064

StdP: 13.63

Time Zone: -6.00 (NAC)

Period: 00-19

WBAN: 94958

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.5	0.6	-11.1	3.2	-3.0	-7.0	4.1	2.0	33.8	29.0	30.8	28.6	7.1	320	0.594

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.5	95.0	72.6	91.3	72.2	88.7	71.5	76.8	88.5	75.1	86.6	73.4	84.9	12.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.2	133.1	83.8	71.7	126.3	82.3	70.0	119.2	80.4	41.8	88.2	40.0	86.5	38.4	85.3	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
28.3	25.0	21.5	-14.5	99.6	5.3	3.8	-18.3	102.3	-21.4	104.5	-24.4	106.6	-28.3	109.3		
			-14.9	80.1	5.2	1.4	-18.7	81.1	-21.7	81.9	-24.6	82.7	-28.4	83.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.5	25.1	26.0	38.6	48.9	59.0	70.1	75.4	72.5	64.4	49.9	37.1	25.8
	DBStd	20.38	11.63	11.98	11.37	9.63	8.94	6.69	5.81	5.68	8.26	9.31	10.28	10.94
	HDD50	3247	772	672	380	131	22	0	0	0	4	116	399	751
	HDD65	6542	1237	1091	820	487	229	28	4	8	110	474	838	1216
	CDD50	3074	0	1	26	99	301	604	788	697	435	112	11	0
	CDD65	896	0	0	1	5	43	182	327	241	91	6	0	0
	CDH74	10086	0	1	30	130	615	2014	3646	2363	1127	153	7	0
	CDH80	4149	0	0	3	30	202	765	1721	973	413	42	0	0
Wind	WSAvg	9.0	9.3	9.7	10.4	11.3	10.1	8.8	7.2	7.0	8.1	8.8	8.9	9.0
Precipitation	PrecAvg	25.2	0.5	0.7	1.3	3.0	3.7	4.3	3.1	3.0	2.2	2.0	0.9	0.6
	PrecMax	35.6	1.5	1.4	3.0	8.7	8.0	10.6	7.8	6.8	4.8	5.6	3.0	3.0
	PrecMin	12.0	0.0	0.0	0.0	0.8	0.2	0.2	0.4	0.7	0.4	0.0	0.0	0.0
	PrecStd	4.5	0.4	0.4	0.9	1.7	2.0	2.3	2.0	1.7	1.2	1.4	0.8	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.5	64.5	77.8	84.4	90.7	95.4	100.3	98.1	93.0	86.0	73.8	58.8
		MCWB	45.3	47.1	55.5	61.0	65.9	71.1	72.7	71.8	68.9	63.3	53.7	44.4
	2%	DB	52.4	56.4	70.6	78.2	85.5	91.2	96.7	92.7	89.3	78.1	66.5	52.2
		MCWB	40.7	44.1	52.8	58.5	65.0	71.6	73.2	73.3	69.4	59.5	50.4	41.8
	5%	DB	46.4	49.7	64.2	72.3	81.1	88.1	92.6	89.0	84.4	72.6	60.5	46.6
		MCWB	37.9	40.3	50.6	56.1	63.6	70.6	73.8	72.9	67.6	56.9	47.9	38.3
	10%	DB	41.6	43.3	58.0	67.2	76.1	84.2	89.4	85.5	81.0	66.9	54.2	41.4
		MCWB	35.3	37.0	47.0	53.2	61.6	68.9	73.0	71.3	65.4	54.3	45.1	35.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.9	49.6	60.0	64.0	70.6	77.1	79.5	78.2	74.1	66.3	56.8	47.5
		MCDB	60.7	63.2	68.2	77.1	83.2	88.6	91.1	89.3	85.9	77.8	67.5	56.1
	2%	WB	41.4	44.3	55.9	60.5	68.0	74.5	77.3	76.2	71.7	62.3	52.4	42.1
		MCDB	51.1	55.7	66.3	73.8	80.5	86.5	89.4	87.3	84.2	72.7	62.8	51.5
	5%	WB	38.7	40.8	51.3	57.5	65.6	72.5	75.7	74.4	69.8	59.1	48.9	38.8
		MCDB	45.8	48.6	63.3	69.7	78.2	84.5	87.5	85.2	81.1	68.6	59.9	45.7
	10%	WB	35.7	37.2	47.1	54.8	62.9	70.7	74.1	72.6	67.7	56.1	45.1	35.4
		MCDB	40.9	43.2	58.4	66.1	74.3	82.1	85.8	82.7	77.6	65.9	53.8	40.7
Mean Daily Temperature Range	5% DB	MDBR	22.2	22.4	24.9	24.4	23.6	23.1	23.5	23.0	26.1	25.9	25.4	21.7
		MCDBR	30.5	32.2	37.0	34.4	30.2	27.1	28.7	27.2	29.8	35.8	36.4	30.0
		MCWBR	20.3	21.0	21.1	18.2	14.4	11.6	11.4	11.8	13.5	19.0	21.6	20.6
	5% WB	MCDBR	28.9	30.6	33.6	29.3	26.2	23.2	24.4	23.4	25.4	28.2	33.7	29.1
		MCWBR	19.8	20.3	20.2	16.3	13.9	11.8	11.8	11.5	12.6	16.5	20.8	20.6
Clear-Sky Solar Irradiance	taub	0.272	0.278	0.303	0.340	0.365	0.380	0.397	0.395	0.358	0.319	0.277	0.268	
	taud	2.478	2.485	2.454	2.366	2.382	2.375	2.360	2.354	2.410	2.499	2.530	2.490	
	Ebn at Noon	283	299	301	295	289	283	277	274	278	279	280	275	
	Edh at Noon	23	27	32	37	38	38	38	37	33	27	22	21	
All-Sky Solar Radiation	RadAvg	680	967	1281	1618	1885	2165	2158	1851	1501	1060	747	580	
	RadStd	59	53	114	84	135	144	83	108	95	111	55	50	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page